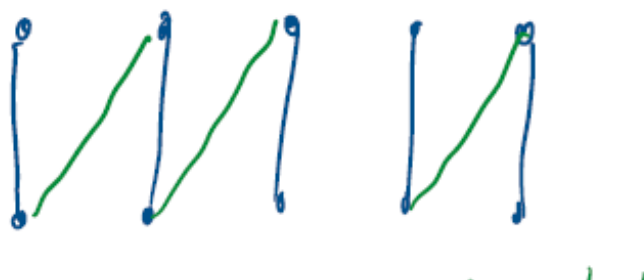


1:50-2:15pm, Monday, April 27, 2026.

1. **Maximal versus Maximum Matchings.** Construct a bipartite graph $G = (X \cup Y, E)$ such that there is a maximal matching of size 3, but the maximum matching has size 5. Show both matchings in your figure.

Answer. Maximal matching = green. Maximum matching = blue.



2. **Symmetric difference of two matchings.** Let M_1 and M_2 be two matchings in an undirected graph such that $|M_1| \neq |M_2|$. Explain why $M_1 \oplus M_2$ cannot be a cycle.

Answer. If $M_1 \oplus M_2$ is a cycle, then the edges of M_1 and M_2 alternate along the cycle, and so $|M_1| = |M_2|$. Since $|M_1| \neq |M_2|$, then $M_1 \oplus M_2$ cannot be a cycle.

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3. **Online unit covering.** Sophia suggests a new algorithm for online unit covering.

OFF-CENTER: If x_i is not covered by any interval in $\mathcal{C}$, add $[x_i - \frac{1}{4}, x_i + \frac{3}{4}]$ to $\mathcal{C}$.

Either show that the competitive ratio of OFF-CENTER is 2, or give an input sequence of real numbers $\sigma = (x_1, x_2, \dots)$ for which $\text{ALG}(\sigma) > 2 \cdot \text{OPT}(\sigma)$.

Answer. Consider $\sigma = (x_1, x_2, x_3) = (0, -0.26, -0.52)$. We have $\text{OPT}=1$, because all three points can be covered by the unit interval $[-1, 0]$.

For $x_1 = 0$, algorithm OFF-CENTER opens the interval $I_1 = [-0.25, 0.75]$.

Since $-0.26 \notin I_1$, then OFF-CENTER opens a new interval $I_2 = [-0.51, 0.49]$.

Since $-0.52 \notin I_1$ and $-0.52 \notin I_2$, OFF-CENTER opens a 3rd interval I_3 .

We see that $\text{ALG}(\sigma) = 3 > 2 = 2 \cdot \text{OPT}(\sigma)$.

4. **Online unit clustering.** Ryan suggests the following algorithm for unit clustering.

GRID: Associate each cluster C_j with an interval $[a_j, a_j + 1]$, where a_j is an integer.

If $x_i \in [a_j, a_j + 1]$ for an existing cluster C_j , then add a_i to cluster C_j ; otherwise open a new clusters $\{x_i\}$ and associate it with the interval $[\lfloor x_i \rfloor, \lfloor x_i \rfloor + 1]$.

Construct an input sequence (x_1, x_2, x_3, x_4) for which GRID uses 4 clusters but $\text{OPT}=2$.

Answer. Consider $(x_1, \dots, x_4) = (0.5, 1.4, 2.3, 3.2)$. Algorithm GRID opens a new cluster for each point, associated with intervals $[0, 1]$, $[1, 2]$, $[2, 3]$ and $[3, 4]$. However, the four points can be arranged into two unit clusters $C_1 = \{0.5, 1.4\}$ and $C_2 = \{2.3, 3.2\}$.